VILNIUS GEDIMINAS TECHNICAL UNIVERSITY

FACULTY OF ELECTRONICS

DEPARTMENT OF ELECTRNICS SYSTEMS

Machine Learning

Practical work 4

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Introduction:

In this practical work I will apply various data preprocessing methods on Adult Census Income Dataset.

Analysis & visualizations:

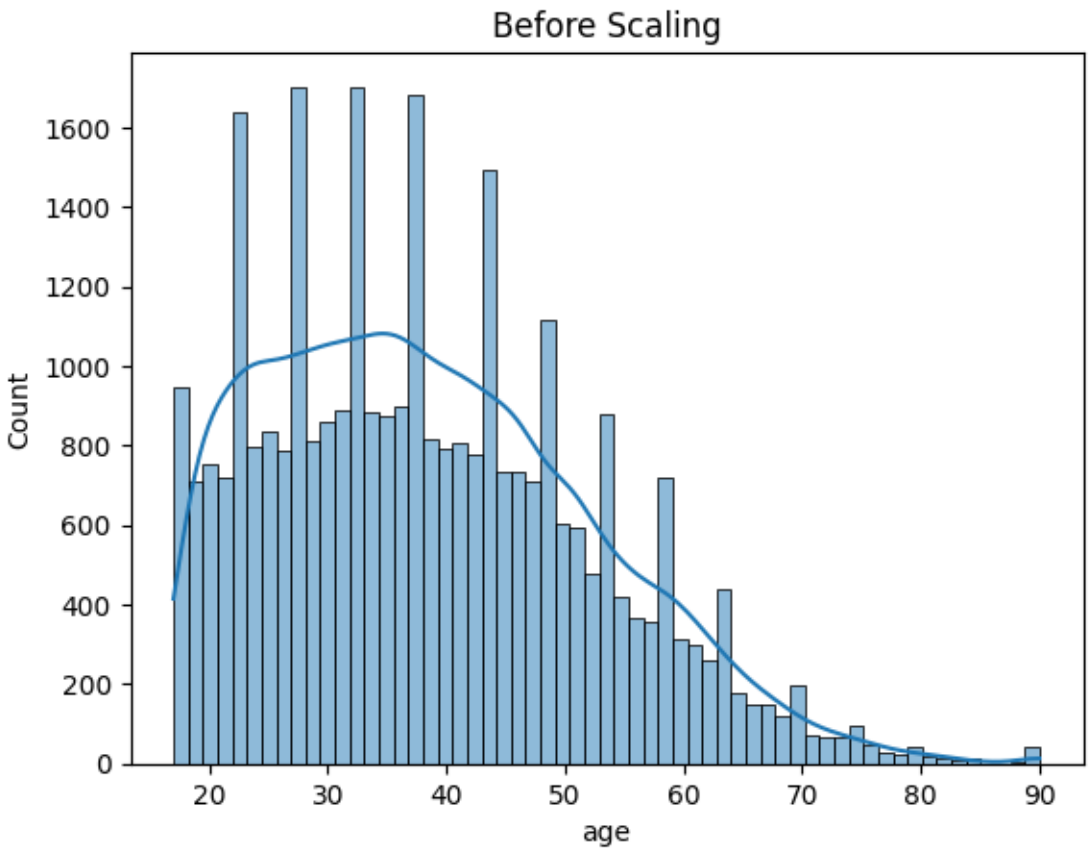
First things first. I found out that there in many cells you can find sign ?, so in order to remove it a raplaced it with NaN. After that I found duplicated in the dataset and dropped them. While looking at the data I noticed that there are two rows that consist mostly of zeros: capital.gain with 92% and capital.gain with 95% of zeros.

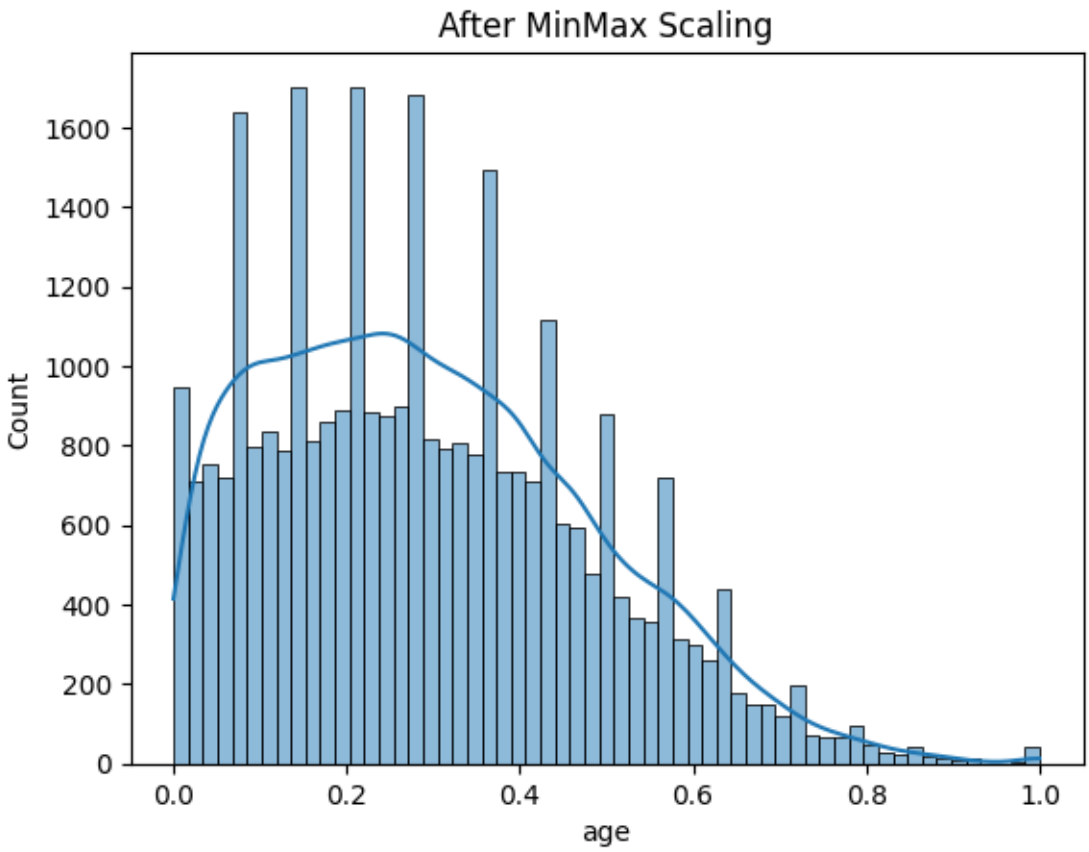
Then I applied different methods of imputation on null data.

```
Missing values before:
 age                 0
workclass         1836
fnlwgt               0
education            0
education.num        0
marital.status       0
occupation        1843
relationship         0
race                 0
sex                  0
hours.per.week       0
native.country     582
income               0
dtype: int64
```

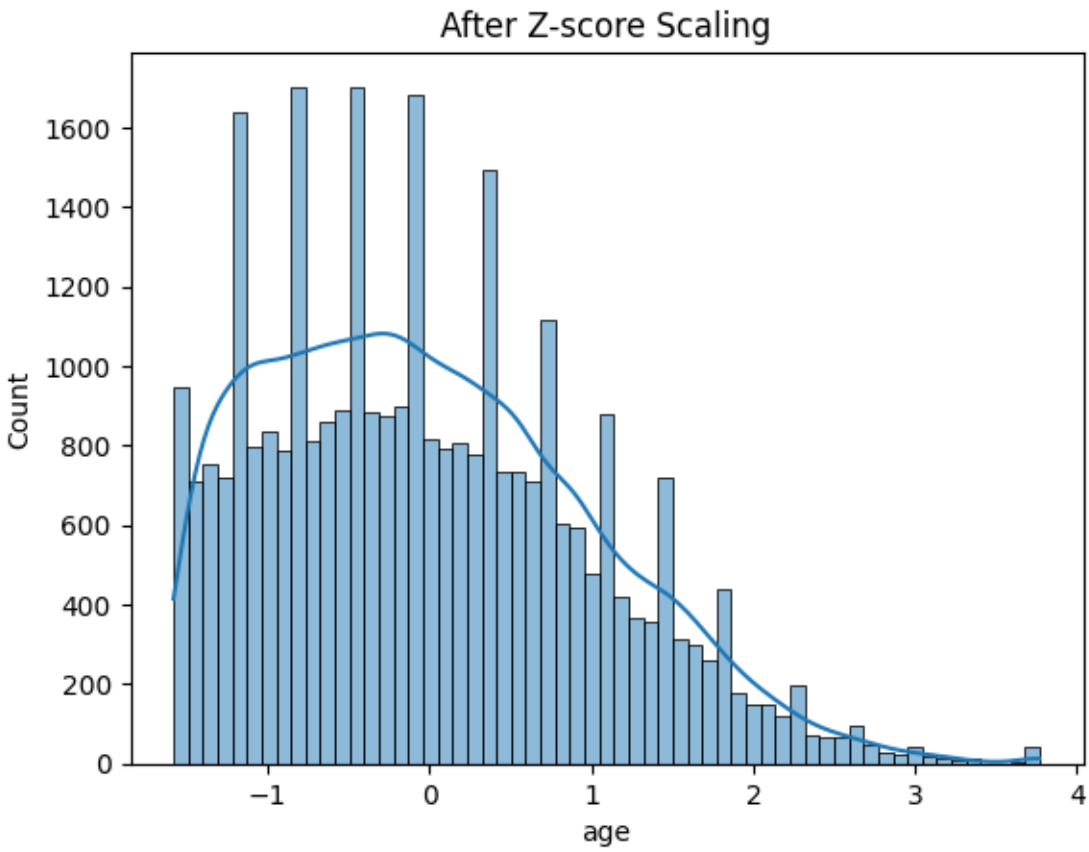
```
Missing after Mean:
 0
Missing after Median:
 0
Missing after KNN:
 0
```

In the next task I had to normalize data using MinMaxScaler and Z-score methods and got the following results:


Before Scaling
Count
age
0
200
400
600
800
1000
1200
1400
1600
20
30
40
50
60
70
80
90

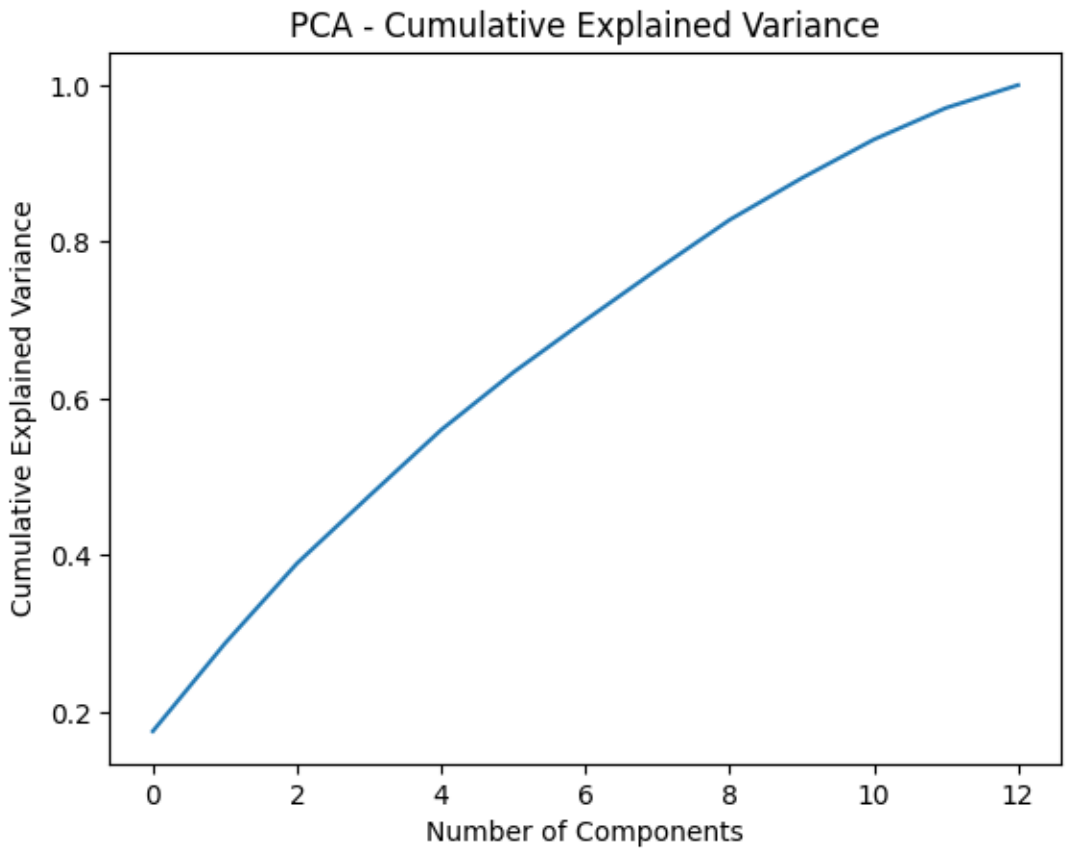


After this method our data is in [0.0, 1.0]



After this method our data is range from -1.21 to 3.77

The next task implies standardizing features using PCA:



This plot shows us how much principal components preserve information.

Then I had to discretize data using equal-width binning, equal-frequency binning and KMeans cluster discretization. For that I specified 5 bins or 5 clusters for KMeans and here are results:

```
Equal Width Bins:
 age
(31.6, 46.2]      12204
(16.927, 31.6]    11446
(46.2, 60.8]       6556
(60.8, 75.4]       2091
(75.4, 90.0]        240
Name: count, dtype: int64

Equal Frequency Bins:
 age
(16.999, 26.0]    7185
(33.0, 41.0]      6760
(50.0, 90.0]      6459
(41.0, 50.0]      6169
(26.0, 33.0]      5964
Name: count, dtype: int64

KMeans Clusters:
 1    9698
2    9403
3    6977
0    4428
4    2031
Name: count, dtype: int64
```

Finally, I made a train test split on the results data, where 20% of which will be used for testing:

```python
target = df_knn.iloc[:, -1]
X = df_knn.iloc[:, :-1]
y = target

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    stratify=y,
    random_state=42
)

print("\nTrain shape:", X_train.shape)
print("Test shape:", X_test.shape)
```

```
Train shape: (26029, 12)
Test shape: (6508, 12)
```

Conclusion:

In this work, the Adult Census Income dataset was preprocessed by handling missing values, removing duplicates, and applying imputation. The data was then normalized using MinMax scaling and Z-score standardization. PCA was used to analyze variance preservation, and discretization methods (equal-width, equal-frequency, and KMeans) were applied. Finally, the dataset was split into training and testing sets (80/20) to prepare it for machine learning tasks.